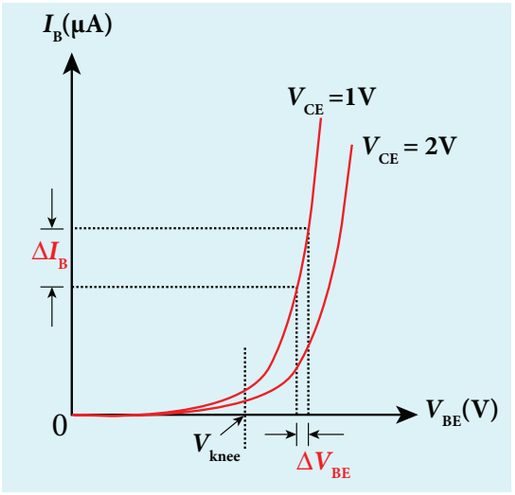


Figure 10.31 Input characteristics

The following observations are made from the graph.

- The curve looks like the forward characteristics of an ordinary *p-n* junction diode.
- There exists a threshold voltage or knee voltage (V_{knee}) below which the base current is very small. This value is 0.7 V for silicon and 0.3 V for germanium transistors. Beyond the knee voltage, the base current increases with the increase in base-emitter voltage.
- It is also noted that the increase in the collector-emitter voltage decreases the base current. This shifts the curve outward. This is because the increase in collector-emitter voltage increases the width of the depletion region which in turn, reduces the effective base width and thereby the base current.

Input impedance

The ratio of the change in base-emitter voltage (ΔV_{BE}) to the corresponding change in base current (ΔI_B) at a constant collector-emitter voltage (V_{CE}) is called the input impedance (r_i). The input impedance is not constant in the lower region of the curve.

$$r_i = \left(\frac{\Delta V_{BE}}{\Delta I_B}\right)_{V_{CE}} \qquad (10.3)$$

The input impedance is high for a transistor in common emitter configuration.

ii) Output characteristics

The output characteristics give the relationship between the collector current (I_C) and the collector to emitter voltage (V_{CE}) at constant input current (I_B) and are shown in Figure 10.32.

Initially, the base current is set to a particular value. Then collector-emitter voltage is increased in suitable steps and the corresponding collector current is recorded. A graph is plotted with V_{CE} along the x-axis and I_C along the y-axis. This procedure is repeated for different values of I_B. The four important regions in the output characteristics are:

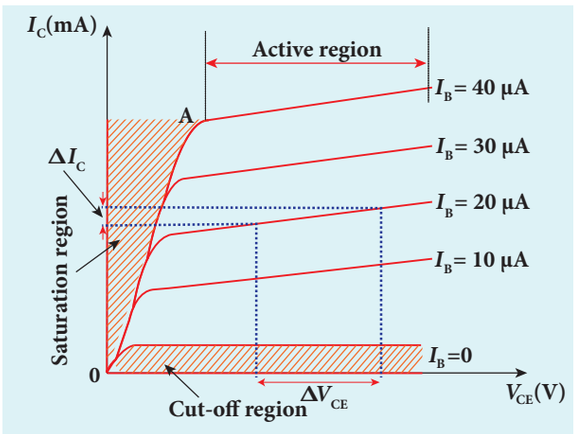


Figure 10.32 Output characteristics

i) *Saturation region*

When V_{CE} is increased above 0 V, the I_C increases rapidly and reaches a saturation value at a particular value of V_{CE}, called knee voltage. The initial part of the curve OA (the ohmic region) between the origin 0 and the knee point A is called saturation region. Transistors are always operated above this knee voltage.

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